FINDING • MHD-WAAM THESIS • PHASE 3

Why the Arc Alone Cannot Melt the Plate

The first arc run, what it proved, and the energy balance that moves the wire inlet onto the critical path

Revision 2 — graded mesh, post Edit 8

waam-arc-01 | 53,856 cells, graded | 756 W half domain | 0.5 s | Al–5%Mg, solidus 815 K, liquidus 906 K

WHAT CHANGED IN REVISION 2

Revision 1 was computed on a 25 x 8 x 9 mm uniform block of 14,400 cells whose boundaries sat well inside the thermal field, and on a solver in which no thermal boundary condition was actually applied. Both are fixed. This edition uses a 60 x 25 x 9 mm graded mesh of 53,856 cells, and Edit 8.

The saturated hot spot moves from 731 K to 832 K. Revision 1 reported the arc falling 84 K short of the solidus; it in fact reaches the solidus and crosses it by 17 K, while still ending 74 K below the liquidus. Anything quoting 431 K of excess or a 731 K peak should be updated.

The conclusion is unchanged. The arc alone produces a 1 mm^3 mushy skin and no liquid whatsoever. The wire inlet, Edit 4, remains on the critical path, and with droplets included the estimated peak rises to ~1066 K rather than the ~922 K of revision 1 — further past the liquidus, not nearer to it.

The boundary-condition anomaly left open at the end of revision 1 is resolved, and was an upstream solver defect. See the closing section.

Revision 1's claim that the magnetic module was switched off was wrong. The switch was absent from the dictionary and the solver defaults to on. No result changes — B = 0 makes the Lorentz force and Joule heating identically zero — but every arc run so far cost 1.79x what it needed to. See "What was run".

THE RESULT IN THREE LINES

The Goldak arc source works: it delivered **99.97%** of its target power, steadily, across 50,000 timesteps, with mass conserved to ten digits. That was what task A1 existed to test, and it passes.

Almost nothing melted. A 1 mm^3 pool against Zhao's ~8 mm wide bead. The reason is not a bug and not a boundary condition: **the arc alone supplies too little power**, because in GMAW a large share of the melting energy arrives as superheated filler droplets, and the wire inlet does not exist yet.

Consequence: Edit 4, the moving wire inlet, was scheduled in Phase 4 as a convenience with no solver work. It is now on the critical path. No weld pool can appear before it exists.

1. What was run

Case	`cases/00-tutorials/waam-arc-01` — bare plate, no filler. An autogenous bead-on-plate pass
Domain	60 x 25 x 9 mm, graded = **53,856** cells, 88 x 34 x 18. 0.5 mm through the pool, expanding by 4 toward the far boundaries; z ungraded, so cell volumes span 16x, not 64x. Half domain, symmetry at y = 0. Substrate 6 mm, argon 3 mm. *Revision 1 used 25 x 8 x 9 mm uniform, 14,400 cells*

Walls	All boundaries adiabatic (`zeroGradient`). Zhao's own condition is convective plus radiative, h = 80 W/m^2/K, which over 0.5 s removes at most ~26 W of 756 W — under 3.5%, and that is a ceiling taken at the peak temperature over the whole face. Recorded as Edit 10, deferred
Arc	Goldak double ellipsoid, a_f 3 mm, a_r 6 mm, b 4 mm, c 4 mm. 1512 W full, **756 W** into the half domain. Travelling at 0.02 m/s
Off	Laser (zero power), vaporisation, arc pressure, wire, shielding gas, plasma drag, welding current
MHD	**ON, and it should not have been.** Revision 1 of this document listed `electromagnetics false` among what was switched off. The key was **absent from the dictionary entirely**, and the solver's internal default is *true*. Every arc run to date, including the one that produced the 832 K result, solved an induction equation and a magnetic divergence-cleaning step at every timestep for a field that is identically zero — **15 of 19 linear solves per step, and two thirds of the elliptic ones**. Measured cost on this case: **1.79x**. With B = 0 the Lorentz force and Joule heating are identically zero, so no result is invalidated; only the runtime was. Now written explicitly as false by the case builder
Duration	0.5 s, 50,000 timesteps (maxDeltaT caps deltaT at 1e-5 s), 4 cores, simple decomposition (4 1 1)

2. What passed

CHECK	RESULT
Arc delivers its target power	**99.97%** of 756 W before renormalisation, constant over the run. The 0.03% shortfall is the Gaussian tail crossing the domain edge. **Re-measured on the 71,808-cell graded mesh: 99.968%**, renormalised x1.00032 to exactly 756 W. The figure carries over unchanged, and a renormalisation of 0.03% means the ellipsoid is essentially fully contained in metal
Both solver patches live	`Metal kappa: two-branch` and `Arc: Goldak double ellipsoid` with f_f = 0.667, f_r = 1.333 summing to 2 as Goldak requires
Laser genuinely idle	`Total rays: 0, Total Q deposited: 0`. At zero power the tracer does not even seed rays, so it costs nothing
Mass conserved	Phase-1 volume fraction 0.6666666666 throughout
Solidus crossed	First cell above the **solidus** at **t = 0.36 s** on the graded mesh. **No cell anywhere reached the 906 K liquidus**, so this is a mushy skin and not liquid — revision 1's phrase 'first liquid' was wrong. Its companion figure, max|U| = 0.256 m/s 'a plausible Marangoni velocity', was also wrong: that maximum was taken over the whole domain and is the buoyant argon, which is already moving at 0.17 m/s at 0.34 s while zero cells are molten. Velocity must be reported per phase, and now is
Free surface flat	Interface Courant identically zero — correct, since arc pressure (Edit 3) does not exist yet

3. The energy balance closes

Summing rho·c_p·(T - 300)·V over every metal cell and comparing against 756·t gives **106.4%** at the end of the run. That excess is the assumed c_p, not a physical one: 1000/1.064 = **940 J/kg/K**, a textbook value for Al-Mg. The balance closes to about zero.

NOTHING IS ESCAPING

Every joule the arc delivers is in the metal. That single fact rules out the entire class of boundary-condition explanations for the small pool — there is no sink to remove.

It also explains a null result that was otherwise baffling: switching the walls from a fixed 300 K sink to perfectly adiabatic changed the answer by **less than one part in 10^8**. If no heat is crossing a boundary, clamping its temperature and insulating it are the same problem. **See section 6 — this is not yet fully explained.**

4. The revision 1 domain was a crucible, not a workpiece — fixed

This section describes the 25 x 8 x 9 mm domain of revision 1 and is kept because it is the reason the mesh was rebuilt. On the graded domain the bulk rise is 34 K/s, not 252 K/s, and the predicted and measured rise over 0.5 s agree at +17 K.

With no losses, all 756 W goes into raising the temperature of 3.18 g of aluminium:

```
756 / (m c_p) = 756 / (0.00318 * 940) = 253 K/s predicted
measured from the run = 252 K/s
```

The *whole plate* is heating uniformly at a quarter of a thousand kelvin per second. Extrapolated, everything melts at once at about t = 2 s. That is a crucible, not a weld. A real substrate is large enough to stay cool and act as a sink, which is what lets a localised pool form at all.

5. The finding: the hot spot saturates at the solidus

Bulk temperature is an artefact of the small domain, so the physically meaningful quantity is **peak minus bulk** — the local excess the arc can sustain above whatever the plate happens to sit at.

These are the graded-mesh numbers. The first edition of this document used a 25 x 8 x 9 mm uniform block whose boundaries sat well inside the thermal field, and its excess column is superseded — see the revision note on page 1. The domain below is 60 x 25 x 9 mm, graded, 53,856 cells, every boundary adiabatic.

TIME	PEAK T	MEAN T	PEAK - MEAN	INCREMENT
0.1 s	702.0 K	303.4 K	398.6 K	—
0.2 s	773.4 K	306.8 K	466.6 K	+68.0 K
0.3 s	806.3 K	310.2 K	496.1 K	+29.5 K
0.4 s	826.4 K	313.6 K	512.8 K	+16.7 K
0.5 s	838.9 K	317.0 K	**521.9 K**	+9.1 K

The excess no longer turns over — that apparent peak-and-decline was the small domain's bulk catching up with its own hot spot. What it does instead is **converge geometrically**: each increment is about 1.85 times smaller than the one before, which is the signature of a diffusion problem settling to steady state. Summing the remaining tail, 9.1 / (1.85 - 1) = 11 K, gives a saturated excess of **about 532 K**.

THE ARC ALONE REACHES THE SOLIDUS AND STOPS THERE

On a cold substrate, where the bulk stays near 300 K, a 532 K excess is a peak of about **832 K**. That is **17 K past the 815 K solidus and 74 K short of the 906 K liquidus**.

The run agrees with the extrapolation: at 0.5 s exactly **8 cells** are above the solidus, 1.00 mm^3 in total, 0.75 mm deep, and **not one cell anywhere in the domain is above the liquidus**. A mushy skin, not a pool.

The conclusion of the first edition survives and the number does not. The margin is far narrower than the 84 K first reported — the arc grazes the solidus rather than falling well short of it — but nothing that can be called a weld pool forms, and no thermal history worth giving to ExaCA comes out of it.

Why the first edition understated the excess. Peak minus bulk is small when the surroundings heat up alongside the hot spot. The old domain was a crucible: its bulk rose at 252 K/s against 34 K/s here, so it climbed toward the peak and compressed the difference. The two runs are worth comparing at 0.1 s, before either had reached its boundaries: the **peaks agree exactly, at 702.0 K**, while the means were already 22 K apart. The bulk error came first and contaminated the peak later — which is why a domain-independence check belongs before the physics conclusions, not after them.

6. What is missing: the droplets

GMAW does not melt the workpiece with the arc alone. It melts it with **superheated filler droplets**, which Zhao's assumption 1 models as hot liquid entering through a virtual tube at 1506 K. That inlet is Edit 4 and does not exist yet.

```
wire 1.2 mm at 0.15 m/s -> 0.450 g/s
droplet enthalpy above 300 K -> 1.49 MJ/kg
  ( c_p*(815-300) + L_f + c_p*(906-815) + c_p*(1506-906) )

droplet power = 0.450e-3 * 1.49e6 = 671 W
arc power = 1512 W
droplets add 44% ON TOP
```

And they do not merely add power. They arrive **already molten at 1506 K** — that energy needs no conduction to become pool, it *is* pool. Scaling the saturated excess by the added power, 532 K x 1.44 = 766 K, lifts the peak from ~832 K to **~1066 K — 160 K past the 906 K liquidus**, not marginally past it.

That scaling is a linear-conduction estimate and it errs in both directions at once. It is an *under*estimate because droplets deposit their enthalpy into a few cells while the arc spreads its own over the whole Goldak ellipsoid. It is an *over*estimate because latent heat, 3.58e5 J/kg, is absorbed isothermally across the mushy range and flattens the temperature rise exactly where the estimate assumes it continues. Which effect wins is a question for the model, not for arithmetic. What the arithmetic settles is the sign: **with droplets there is a pool, without them there is not**, and the gap is not close enough for the caveats to change the answer.

7. Consequences for the plan

	CHANGE
1	**Edit 4 moves onto the critical path.** It was scheduled in Phase 4 as a dictionary-only convenience. Nothing downstream can be validated without it, because no weld pool exists until the wire does.

	CHANGE
2	**The A1 gate was mis-specified.** 'A pool forms' was written into the acceptance criteria for a model that physically cannot form one. The correct A1 gate is the power-delivery check, which passed.
3	**Domain sizing must be revisited before A2. DONE.** The domain is now 60 x 25 x 9 mm on a graded mesh, 53,856 cells, 0.5 mm through the pool and expanding by 4 toward the far boundaries. It is verified independent of the far-field condition: the outermost cell layer in y sits at **300.2 K** at 0.5 s, 0.04% of the peak rise, so no choice of condition there can matter over this run length. Bulk heating fell from 252 K/s to 34 K/s, matching prediction to 0 K.
4	**The Goldak parameters deserve a check against the paper.** c = 4 mm is the depth semi-axis for a pool around 2 mm deep. A smaller c concentrates the heat and raises the peak, and the current values came from the case matrix rather than a verified reading of Zhao Table 4.

RESOLVED: THE BOUNDARY CONDITION ANOMALY WAS AN UPSTREAM DEFECT

The anomaly was real and the reasoning that flagged it was right: a bottom cell layer at 524.8 K with a 300 K clamp 0.25 mm away implies a flux far larger than the arc input, so the two runs could not have matched. They matched because **no thermal boundary condition in the solver was doing anything at all**.

The proposed test was run — walls at `fixedValue 2000` — and the bottom layer did not respond. The cause is in `createFields.H`: `kappa` and `cp` are constructed with a `dimensionedScalar`, which gives every patch a `calculated` boundary field initialised to zero, and `calculated` has no evaluation rule, so the `correctBoundaryConditions()` call in `updateKappaCp.H` is a no-op. Both transport routes in `TEqn.H` then vanish at the boundary: `laplacian(kappa, T)` because kappa_f = 0, and `div(rhophicp, T)` because it interpolates the same zeroed cp. The domain was a perfectly insulated box whatever the dictionaries said.

Fixed as **Edit 8** by passing `zeroGradientFvPatchScalarField::typeName` to both constructors. Verified: the 2000 K wall test moved the bottom layer from 313.1 K to 1969.8 K. **Every wall-thermal-BC result taken before Edit 8 is void**, and the planned wall-BC sensitivity study would have reported no sensitivity — the same class of silent false negative as the `sigma_e_gas_ratio` bug in Phase 2.